DTP

Debug and Test Ports

Beta release | Open-source RTL | Apache 2.0

HIGHLIGHTS

- JTAG TAP and boundary-scan access implementing IEEE 1149.1-2013; the boundary-scan network and logic are supplied by adopter DFT tooling and pad IP
- iJTAG instrument networks implementing IEEE 1687-2014
- Configurable secondary TAP topology, compliant with IEEE 1838, for local and downstream chiplet access
- JTAG-to-AXI paths for SMC fabric, SMC OTP, and SEP OTP debug access
- Cross-trigger routing between chiplet-local interfaces and external cross trigger ports (CTPs)
- Lifecycle-controlled disable inputs for all internal and external debug access ports
- Integrated debug clock-stop, boot-stall, and multi-target reset controls

SYSTEM ROLE

DTP sits between external debug and test equipment, OCAH management and security functions, adopter IP, and neighboring chiplets. It is a component of the OCAH SMU, a first-class subsystem delivered as part of OCAH whose reference components include the subsystem wrappers and IP-integration modules. DTP can also form a subsystem-level boundary when the adopter supplies its lifecycle policy, fabric, reset, clock, scan, and pad connections.

OVERVIEW

The Debug and Test Ports (DTP) subsystem combines a JTAG Interface Unit based on IEEE 1149.1-2013, IEEE 1687-2014, and IEEE 1838 with a Cross Trigger Network for local and inter-chiplet triggers, boot, reset, and clock control. Optional JTAG-to-AXI bridges access the SMC fabric and SMC/SEP OTP interfaces; adopter-IP access depends on the downstream address map.

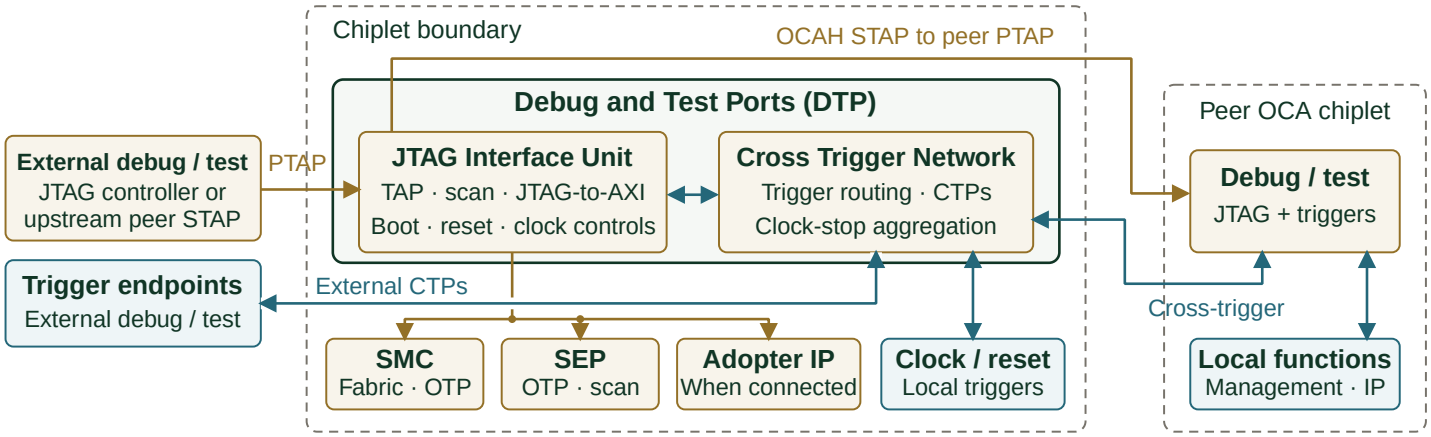
Active-high disable inputs are synchronized into the scan domain and gate the corresponding paths. In an SMU with SEP, SEP supplies the lifecycle-control policy. Without SEP, debug and test paths are open by default; protection requires routing and gating all such access through a primary chiplet's SEP-controlled DTP path. For standalone DTP, the adopter supplies the policy source and permitted address ranges.

AT A GLANCE

Feature	Description
External / internal triggers	1-32 CTPs / 0-32 internal interfaces
Clock-stop request inputs	1-32
Cross-trigger CSR port	32-bit address, 32-bit data AXI4-Lite
SMC debug manager	56-bit address, 64-bit data AXI4
OTP debug managers	32-bit address, 32-bit data AXI4-Lite
Primary instruction register	6 bits

The reference defaults are 16 CTPs, 10 internal trigger interfaces, and nine clock-stop request inputs. The default SMU uses two of the internal trigger interfaces and one of the clock-stop request inputs for SMC connections, so its external boundary exposes eight internal trigger interfaces and eight clock-stop request inputs; all 16 CTPs remain externally visible.

SYSTEM CONTEXT



Adopter supplies pads, policy, clocks, resets, scan chains and permitted mappings.

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CAPABILITIES

JTAG and scan

- Primary TAP with IDCODE and BYPASS; SAMPLE/PRELOAD and EXTEST are compile-time optional with boundary scan enabled
- Compile-time optional INTEST, CLAMP, HIGHZ, RUNBIST, EXTEST_PULSE, EXTEST_TRAIN, TMP, and CLAMP hold/release functions
- Boundary-scan host interface plus separate DFD, secure DFT, non-secure DFT, extended STAP, I/O STAP, SMC STAP, SEP STAP, and extra-STAP connections
- Capability registers report enabled functions and JTAG-to-AXI limits

Debug and control

- Single and series JTAG-to-AXI transactions, including incrementing and fixed-address series operations and target error reporting
- Independent lifecycle disable fields for three bridge paths and eight scan-side paths
- Independently enabled reset-control outputs for SMC, SEP, and adopter logic
- Debugger-controlled boot-stall and functional clock-stop requests

Cross trigger

- Bidirectional external CTP signaling and chiplet-local trigger interfaces
- Software-controlled source/destination routing with point-to-point and wire-OR behavior
- Pulse synchronization or four-phase request/acknowledge mode per internal trigger interface

INTERFACES AND CONFIGURATION

Feature	Integration options
Clocks and resets	System clock/reset, JTAG TCK/TRST_N, JTAG power-on reset, scan reset, test enable
Lifecycle policy	Active-high, per-path debug and test disable controls
Scan	PTAP pins; BSR/iJTAG scan controls; I/O, SMC, SEP, and extra STAPs
Fabric	One AXI4 manager, two AXI4-Lite managers, one AXI4-Lite subordinate
Cross trigger	External pad controls, internal source/sink interfaces, clock-stop requests and output
Build-time	Instruction/path enables, extra STAP count, IDCODE/version fields, reset-control output types, trigger protocol modes, AXI type parameters

DOCUMENT CONTROL

Version	Date	Author	Change

INTEGRATION DEPENDENCIES

- Generate lifecycle debug-disable policy outside DTP and connect the current active-high debug-disable contract.
- Replace the zero-valued IDCODE manufacturer, part, revision, and OCAH version defaults with release-specific values.
- Provide safe clock, reset, CDC, pad, and clock-gating integration, and establish project-specific TCK constraints.
- Configure the SMC, SEP, and adopter reset-control output types and enables, and connect or correctly bypass every enabled scan and secondary-TAP chain.
- Connect enabled AXI manager ports only to address maps permitted by the product security policy. Disabled and unused interfaces require documented tie-offs.
- Integrate the cross-trigger CSR subordinate into the management address map and connect the required local, external, and clock-stop endpoints.

VERIFICATION AND MATURITY

The OCAH verification dashboard publishes the current verification and maturity status. The repository's open-source PyUVM environment exercises TAP operation, standard and optional instructions, JTAG-to-AXI data and error paths, backpressure/reset cases, iJTAG and STAP routing, cross-trigger modes, clock controls, and lifecycle-disable matrices. Scoreboards, reference models, coverage intent, and deliberately corrupted-checker tests guard against vacuous passes.

This beta sheet does not specify a characterized operating frequency, and no third-party standards certification, silicon characterization, or portable frequency, area, and power figures are claimed. Those results depend on the selected configuration, libraries, constraints, physical implementation, and target process.

DELIVERABLES

- SystemVerilog RTL and dependency filelists
- SystemRDL-derived cross-trigger register collateral
- Integration, architecture, interface, and verification-plan documentation
- Open-source cocotb/PyUVM tests, reusable JTAG/AXI agents, scoreboards, reference models, and functional coverage
- Lint collateral and lint, CDC, and RDC waivers
- Synthesis and timing-constraint starting points
- Apache License 2.0 repository distribution

RESOURCES

Documentation: docs.tenstorrent.com/tt-oca-harness
DTP source documentation: `hw/sys/dtp/doc/`
RTL: `hw/sys/dtp/rtl/`
Verification: `hw/sys/dtp/dv/`
Synthesis collateral: `hw/sys/dtp/synth/`
Integrator Guide: `doc/integrator/`
Project: github.com/tenstorrent/tt-oca-harness

TERMS

BSR: boundary-scan register. CSR: control and status register. CTP: cross trigger port. DFD: design-for-debug. DFT: design for test. iJTAG: internal JTAG (IEEE 1687). OTP: one-time programmable. PTAP: primary TAP. SEP: security processor. SMC: system management controller. SMU: system management unit. STAP: secondary TAP. TMP: test mode persistence.

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